



Research Paper

Unlocking spatiotemporal insights in crack cocaine use in Europe through a wastewater-based epidemiology approach

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Abstract

Background and aims

The increasing prevalence of crack cocaine across Europe is concerning. In contrast to benzoylecgonine (BE), the analysis of anhydroecgonine methyl ester (AEME) in influent wastewater (IWW) supports the estimation of crack cocaine use at the population level, since it is exclusively formed as a pyrolysis product during inhalation. This study aims to monitor spatiotemporal trends in the population-normalized mass loads of AEME and BE in IWW of different European communities, as a proxy for human consumption.

Methods

A validated analytical method quantified AEME and BE in IWW from different European cities. A 2024 spatial analysis compared Western and Eastern European cities, while monthly IWW composites in May 2023 covered 24 cities in Flanders (Belgium). Yearly trends were evaluated over six years (2019–2024) in Antwerp and Brussels.

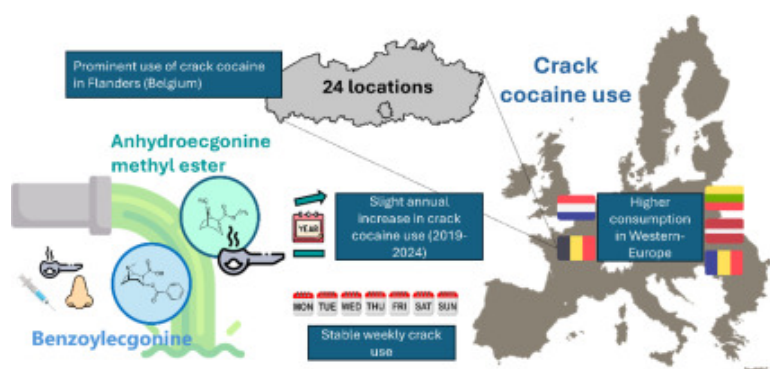
Results

Higher population-normalized mass loads of AEME and BE were observed in Belgium and the Netherlands compared to Eastern European countries. In Lithuania, Latvia and Romania, AEME concentrations were generally below the lower limit of quantification. The spatial analysis in Flanders showed that crack cocaine use was not limited to urban locations. In Belgium, the population-normalized mass loads of AEME remained stable throughout the six-year period with no clear week-weekend pattern, suggesting continuous use due to its highly addictive properties.

Conclusions

AEME is a reliable biomarker for estimating crack cocaine use in populations using IWW analysis. WBE is a strong and effective monitoring tool, essential to feed situational analysis and allows timely insights to support public health surveillance.

Graphical abstract



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Introduction

Cocaine, derived from the South American *Erythroxylon coca*, was originally used medically but became an expensive drug used in recreational settings in the 1970s. Cocaine is Europe's most used stimulant and its availability is increasing (EUDA, 2025a). Cocaine can be snorted as a powdered salt (cocaine HCl) and inhaled in its base form (crack cocaine) (EUDA, 2025a; EUROPOL and EUDA, 2024; Fryer et al., 2013). The use of crack cocaine is more concerning compared to cocaine HCl because of its intense highs that wear off quickly, leading to strong cravings making it highly addictive (EUDA, 2025a; EUROPOL and EUDA, 2024; Fryer et al., 2013). Its use is associated with strong dependence, numerous hospital emergencies, overdose deaths, health risks, but also social consequences such as unstable home environments and elevated crime rates (EUDA, 2024a, 2024b; Fryer et al., 2013; Morelos-Santana et al., 2024; UNODC, 2021).

In the 2024 European Web Survey on Drugs, 29% of respondents in the European Union or Norway reported having consumed cocaine HCl, crack cocaine or both in the last year (EUDA, 2025b). In Europe, cocaine ranked as the second most frequently reported problem drug among people entering specialist drug treatment for the first time in 2023. Additionally, the number of first-time treatment entrants who reported crack cocaine as primary drug increased by 35% between 2018 and 2023 (EUDA, 2025a). While traditional epidemiological data sources, such as surveys, provide useful information about drug use patterns and context, these might not be best fit for prevalence estimation as they do not include all demographic groups, resulting in samples that are not representative for the entire population. In addition, objective measurements of highly stigmatised and hidden behaviour is complicated, resulting in various biases in the collected self-reported figures (EUDA, 2025b; Gisle and Maetens, 2025). In particular, surveys often struggle to reach people who use crack, whose socio-economic characteristics can pose significant barriers. Crack cocaine use is more stigmatized than powder cocaine use, since it is often associated with more marginalized population groups (Bawin et al., 2025). Finally, the interpretation of seizure data is also complicated since the majority of users prepare crack cocaine from cocaine HCl using “free-basing” approaches (Bawin et al., 2025) so seized crack cocaine is likely not representative for its consumption volume. However, the market for premade crack cocaine is increasing (Bawin et al., 2025). In the past year, syringe exchange programs and drug consumption rooms also distributed sterile crack pipes, serving as the main source of contextual information and proxy for crack cocaine use. However, their coverage is limited and uneven across the country, complicating population-level measurements (EUDA, 2025c).

Wastewater-based epidemiology (WBE) helps tracking stimulant use at the population level by measuring specific human biomarkers in influent wastewater samples (IWW) (Bade et al., 2023, 2021; Baz-Lomba et al., 2025; Burgard et al., 2019; González-Mariño et al., 2020; Gunnar and Kankaanpää, 2019; Mackie et al., 2019). To enable spatiotemporal comparisons, measured biomarker concentrations are transformed in population-normalized mass loads (mg/day/1000 inhabitants) by multiplying with daily wastewater flow rates and dividing by the population number served by the wastewater treatment plant. These population-normalized mass loads serve as a proxy for human consumption. To ensure accurate measurement of human biomarkers, they should be stable in IWW (i.e. no biotransformation or loss due to sorption to biofilms or solids), specific to human metabolism, and sufficiently excreted to be measurable (Castiglioni et al., 2013; Pico and Barcelo, 2021). Thanks to its high spatio-temporal resolution, WBE has been used successfully to monitor spatiotemporal trends in the population-normalized mass loads of emerging drugs, such as fentanyl analogues, ketamine and new psychoactive substances (Bade et al., 2023; Boogaerts et al., 2021; González-Mariño et al., 2020; Hahn et al., 2022; O’Keeffe, 2021; Salgueiro-Gonzalez et al., 2024; Steenbeek et al., 2022).

Consumed cocaine, either snorted powder cocaine or inhaled crack cocaine mainly hydrolyses to ecgonine methyl ester and benzoylecgonine (BE) (See Fig. S1), which can further hydrolyse to ecgonine. As a result, the majority of IWW monitoring campaigns are using BE and cannot

distinguish between cocaine HCl and crack cocaine use. However, when crack cocaine is heated for inhalation, it also forms specific pyrolytic products such as anhydroecgonine methyl ester (AEME), which is further transformed to anhydroecgonine. AEME is the main pyrolysis product of crack cocaine and can therefore be considered as a specific biomarker (Gomes et al., 2018; González-Mariño et al., 2019; Jones, 2017; Randall, 1978). Additionally, the bioavailability of cocaine depends on the route of administration, being 85% when cocaine is snorted and 33–77% when inhaled, resulting in different excretion patterns. When cocaine is inhaled, only about one-third of the initial dose is inhaled, while a larger amount of AEME is produced, making it a suitable for crack cocaine use (Devault et al., 2018; Randall, 1978). To our knowledge, Bisceglia et al., González-Marino et al., Steenbeek et al. and Devault et al. are the only studies that have previously provided insights on crack biomarkers in IWW and possible use of AEME as a biomarker, highlighting the need for additional data (Bisceglia et al., 2010; Devault et al., 2018; González-Mariño et al., 2019; Steenbeek et al., 2022). *Steenbeek et al. 2022* previously studied the in-sample stability of crack cocaine biomarkers, including AEME (Steenbeek et al., 2022). The latter remained stable in IWW for up to 96 h, allowing it to track crack cocaine use in WBE.

This study aims to monitor spatiotemporal trends in the population-normalized mass loads of AEME and BE in IWW of different European communities, as a proxy for (crack) cocaine consumption. To achieve this, an analytical method was validated and applied to >200 IWW samples across a multitude of European locations and time periods. A broad spatial screening was executed in different European cities to highlight geographical differences in the population-normalized mass loads of AEME and BE. Additionally, a more detailed spatial analysis was performed across 24 Belgian cities to highlight regional differences in their population-normalized mass loads and assess the use of this illicit drug nationally. Furthermore, temporal trends in the population-normalized mass loads of these cocaine biomarkers were investigated. Daily variations in the population-normalized mass loads of AEME and BE were monitored over the course of a week to evaluate week-weekend patterns. A six-year temporal analysis was performed to assess yearly trends in the population-normalized mass loads of AEME in the wastewater treatment plants (WWTP) Antwerp-South and Brussels-North to evaluate the emergence of crack cocaine in Belgium.

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Section snippets

Materials and biomarker selection

A detailed description of the used materials, reagents, and data processing can be found in Section S.1 of the Supplementary Information (SI). In this study, spatiotemporal trends in the population-normalized mass loads of AEME and BE were investigated. BE and AEME have been proven to be stable in wastewater samples (Castiglioni et al., 2013; Steenbeek et al., 2022). ...

Sampling

Daily 24-hour composite IWW samples were collected from 33 European locations, including 24 Belgian cities. Samples were acquired ...

Method validation

The performance criteria (e.g., selectivity, specificity, carry-over, accuracy, precision, recovery, intra- and inter-day accuracy, intra- and inter-day precision) were successfully achieved for both biomarkers (see also Fig. S3), making the method suitable for its intended application. A linear calibration range was obtained for both AEME and BE with lower limit of quantifications of 6.5 ng/L and 50 ng/L, respectively. The calibration range per biomarker is listed in Table S7. Process ...

Spatial trends in Europe

The concentrations of BE were above the lower limit of quantification in all study locations. The population-normalized mass loads of BE were higher in Western European countries compared to Eastern Europe (see Fig. S5) with the highest population-normalized mass loads found in Antwerp (on average 1927 ± 482 mg/day/1000 inh), Amsterdam (1672 ± 101 mg/day/1000 inh) and Rotterdam (1303 ± 376 mg/day/1000 inh). Significant differences in the population-normalized mass loads of BE were found between ...

Conclusions

The results of this study show that AEME in IWW can be used as an effective biomarker to measure crack cocaine consumption at the population level. Since WBE is one of the few monitoring systems that can distinguish between the use of cocaine HCl and crack cocaine, future WBE monitoring campaigns (e.g. SCORE-EUDA monitoring campaign) should consider the inclusion of AEME alongside BE to better differentiate between these forms of cocaine use. This is especially important since robust population ...

CRedit authorship contribution statement

Imane Taamarti: Writing – review & editing, Writing – original draft, Visualization, Validation, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Natan Van Wichelen:** Writing – review & editing, Writing – original draft, Validation, Methodology,

Investigation, Formal analysis, Conceptualization. **Maarten Quireyns**: Writing – review & editing, Writing – original draft, Visualization, Methodology, Investigation, Formal analysis, Conceptualization. **Thomas ter Laak**: ...

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper. ...

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